

COS 433: Cryptography

Announcements:

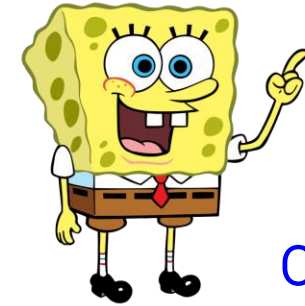
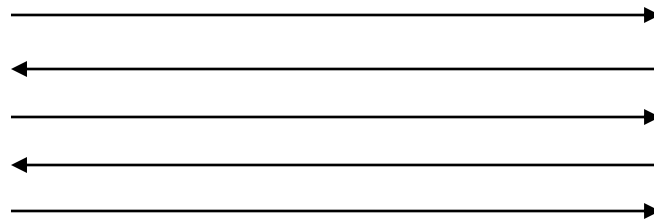
- Problem Set 5 due 12/3
- **Final problem set** released 12/1, due **Sunday 12/14 11:59pm**
 - No collaboration allowed
 - You may use anything available on our course website (including the lecture slides and textbooks)
 - No other sources may be consulted
 - No late submissions (due date assigned by the university)

Proof Systems

Claim: $x \in L$



$P(x, w)$



$V(x)$

Output: $b \in \{0,1\}$

Definition: a proof system for a language $L \subset \{0,1\}^*$ consists of a pair of interactive algorithms (P, V) run on an input $x \in \{0,1\}^n$

- V will always be a **poly**(n)-time algorithm
- P can be inefficient, or **poly**(n)-time given a witness w

It must satisfy the following properties:

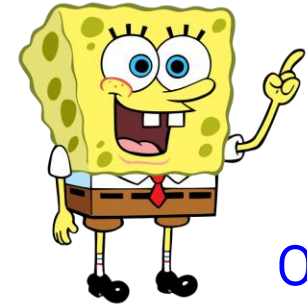
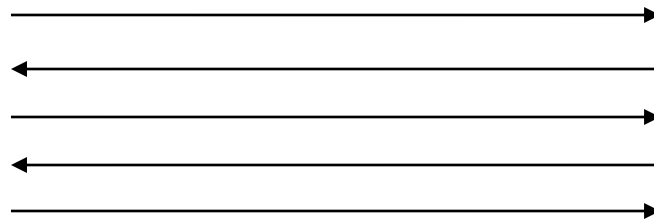
1. **Completeness:** if $x \in L$ and the two parties run (P, V) , then V outputs 1 (accept)
2. **Soundness:** if $x \notin L$, then for **any** strategy P^* , V accepts with probability **negl**(n).

Honest-Verifier Zero Knowledge

Claim: $x \in L$



$P(x, w)$



$V(x)$

Output: $b \in \{0,1\}$

Definition: a proof system (P, V) for a language L is **honest-verifier zero knowledge** if there exists a **polynomial-time simulator**

$$\text{Sim}(x) \rightarrow \tau$$

such that **for all $x \in L$** and all valid w ,

$$\text{Sim}(x) \approx_c \text{view}_V(x, w)$$

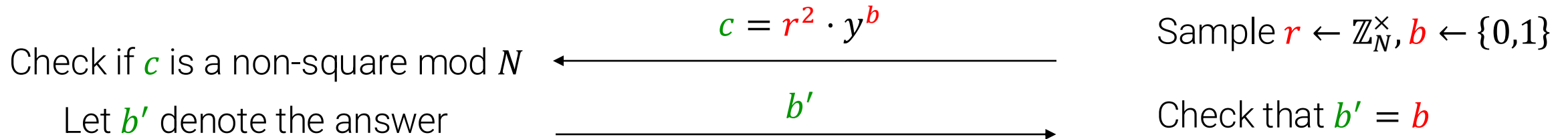
Simulation paradigm: everything V learned, V could have generated on their own!

Proof Systems: Quadratic Non-Residuosity

$P(p, q, y)$

Claim: $y \notin \text{QR}_N$

$V(N, y)$



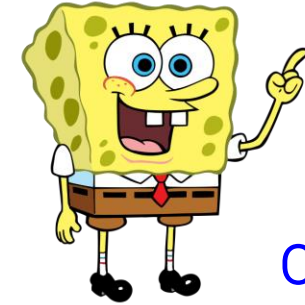
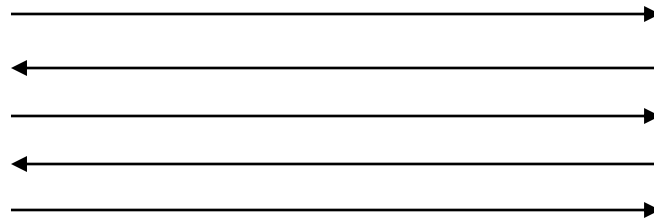
Problem: what if we don't trust V to behave according to the protocol specification?

Malicious-Verifier Zero Knowledge

Claim: $x \in L$



$P(x, w)$



$V(x)$

Output: $b \in \{0,1\}$

Definition: a proof system (P, V) for a language L is malicious verifier zero knowledge if for all polynomial-time V^* , there exists a polynomial-time simulator

$$\text{Sim}^*(x) \rightarrow \tau$$

such that for all $x \in L$ and all valid w ,

$$\text{Sim}^*(x) \approx_c \text{view}_{V^*}(x, w)$$

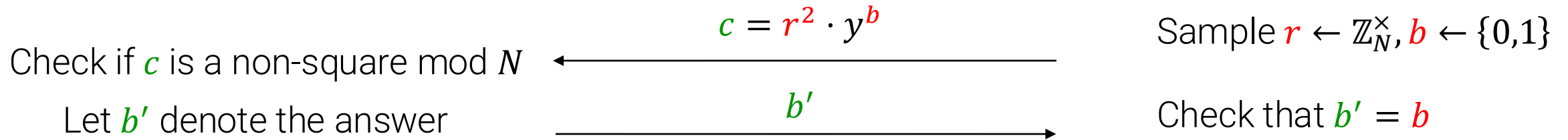
Simulation paradigm: everything V^* learned, V^* could have generated on their own!

Proof Systems: Quadratic Non-Residuosity

$P(p, q, y)$

Claim: $y \notin \text{QR}_N$

$V(N, y)$



Is this protocol malicious-verifier ZK?

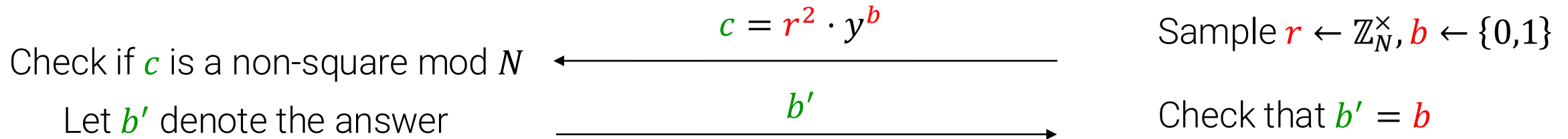
No!

Proof Systems: Quadratic Non-Residuosity

$P(p, q, y)$

Claim: $y \notin \text{QR}_N$

$V(N, y)$



Is this protocol malicious-verifier ZK?

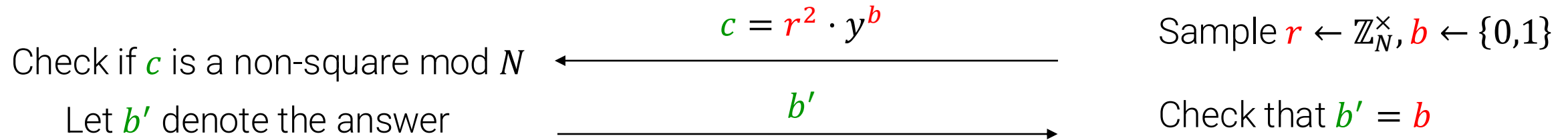
- Intuition: malicious V^* could send c to be any number mod N , where V^* does not know if c is a square modulo N
- Honest P will tell V^* this information, so V^* has learned something

Proof Systems: Quadratic Non-Residuosity

$P(p, q, y)$

Claim: $y \notin \text{QR}_N$

$V(N, y)$



Magic: there's a way to change this protocol, to make it fully ZK!

A little too complicated; we will switch to a different example.

Zero-Knowledge Proof for QR

$P(N, y, w)$

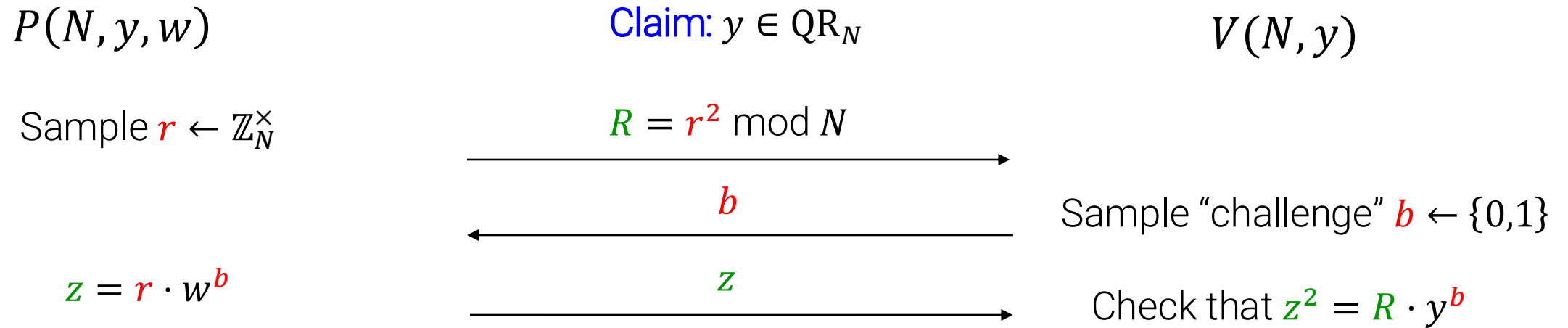
Claim: $y \in \text{QR}_N$

$V(N, y)$

Example: quadratic residuosity

- $L_{\text{QR}} = \{(N, y): y \text{ is a square mod } N\}$
- Witness w : $w^2 \equiv y \pmod{N}$

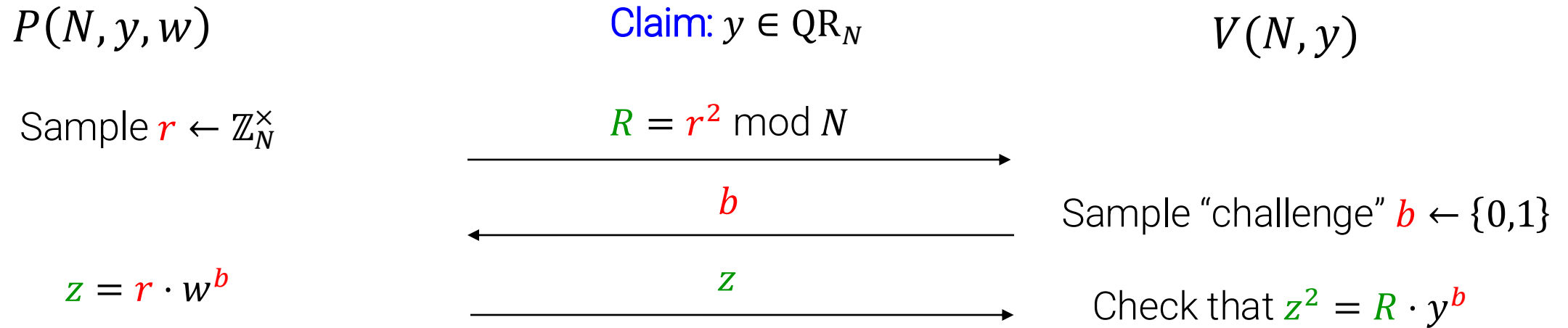
Zero-Knowledge Proof for QR



Intuition: “I’ll show you that I could send you a square root if I wanted... but I won’t!”

“I know a square root of R and a square root of $R \cdot y$, but you only get to see one”

Zero-Knowledge Proof for QR

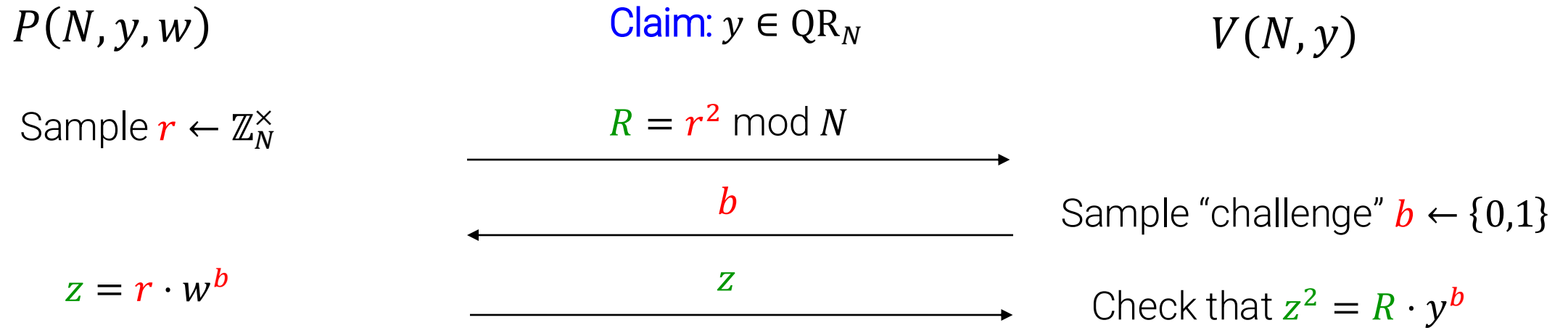


Completeness: $(r \cdot w^b)^2 \equiv R \cdot y^b$

Soundness: if y is **not** a square, then **impossible** to have answers to both challenges!

- For any $R \in \mathbb{Z}_N^\times$, either R is not a square or $R \cdot y$ is not a square
- For any P^* , V rejects with probability at least $\frac{1}{2}$
- Repeat the protocol (sequentially!) to improve the soundness error.

Zero-Knowledge Proof for QR



Claim: this proof system is HVZK

Proof: define the simulator

$$\text{Sim}(N, y) = \left((N, y), \overbrace{b, s^2 \cdot y^{-b}}^R, \underbrace{s}_z \right)$$

If $y \equiv w^2$, the output of $\text{Sim}(N, y)$ is **identical** to $\text{view}_V(N, y, w)$

Zero-Knowledge Proof for QR

$P(N, y, w)$

Claim: $y \in \text{QR}_N$

$V(N, y)$

Sample $r \leftarrow \mathbb{Z}_N^\times$

$$R = r^2 \bmod N$$

b

Sample "challenge" $b \leftarrow \{0,1\}$

$$z = r \cdot w^b$$

z

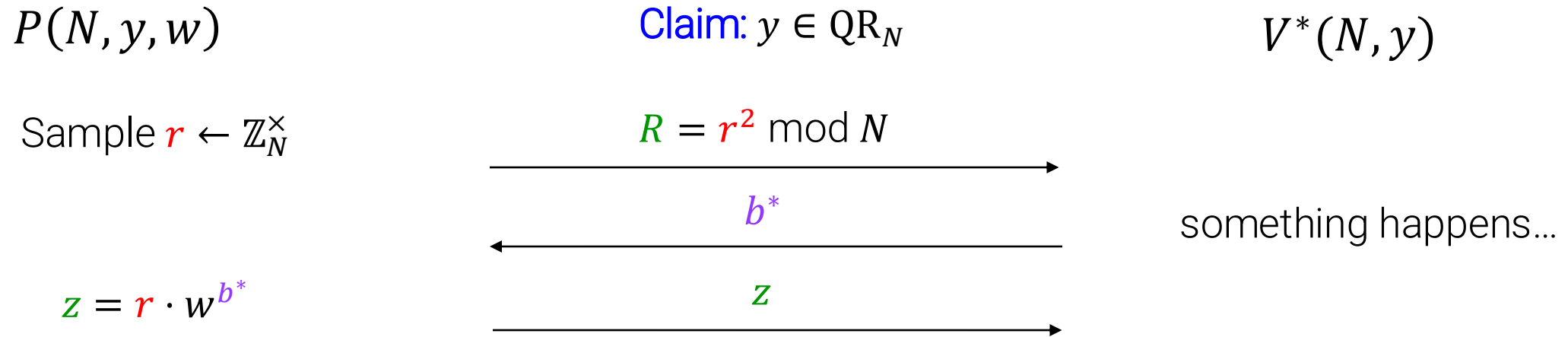
Check that $z^2 = R \cdot y^b$

Claim: this proof system is ZK

Proof: ???

Any ideas?

Zero-Knowledge Proof for QR



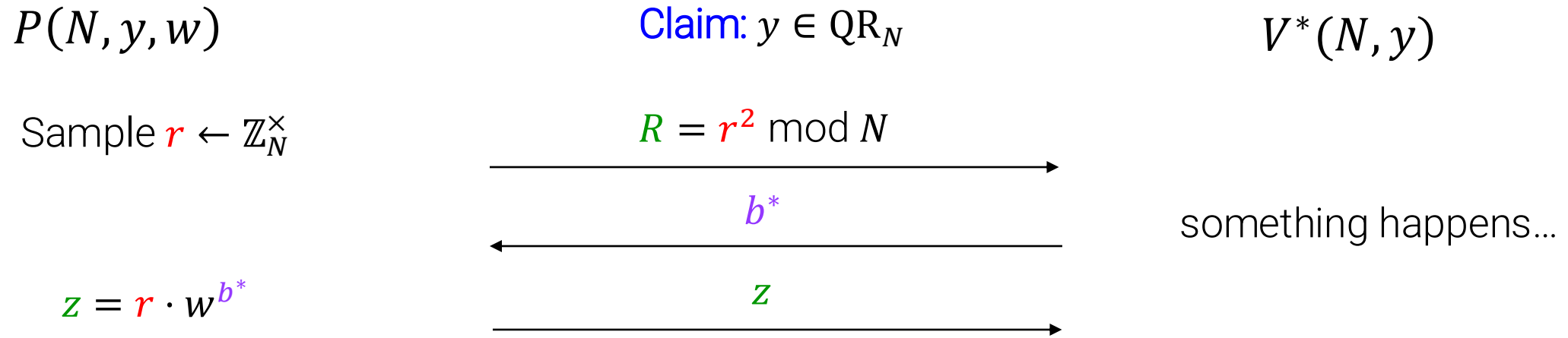
Claim: this proof system is ZK

Proof: Given V^* , need to define Sim^*

Can we use $\text{Sim}^* = \text{Sim}$?

$$\text{Sim}(N, y) = \left((N, y), b, s^2 \cdot y^{-b}, s \right)$$

Zero-Knowledge Proof for QR



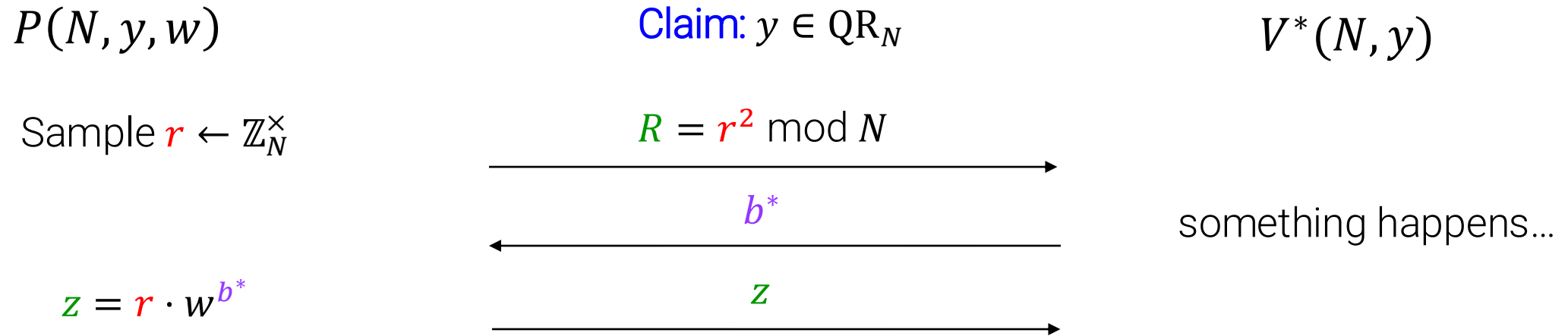
Claim: this proof system is ZK

Proof: Given V^* , need to define Sim^*

Can we use $\text{Sim}^* = \text{Sim}$? No! What if $V^*(R) = 0$ for all R ?

$$\text{Sim}^*(N, y) = ((N, y), 0, s^2, s)$$

Zero-Knowledge Proof for QR



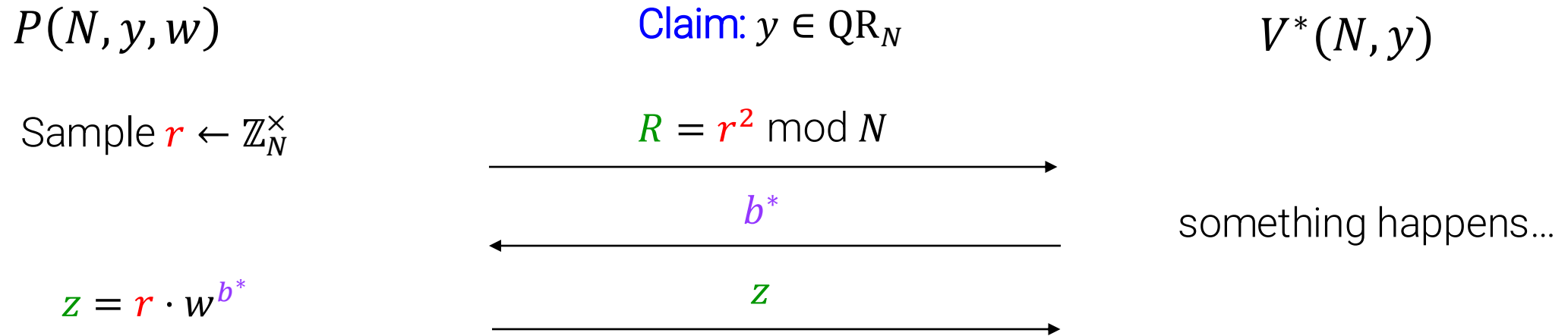
Claim: this proof system is ZK

Proof: Given V^* , need to define Sim^*

Can we use $\text{Sim}^* = \text{Sim}$? No! What if $V^*(R) = 1$ for all R ?

$$\text{Sim}^*(N, y) = ((N, y), 1, s^2 \cdot y^{-1}, s)$$

Zero-Knowledge Proof for QR



Claim: this proof system is ZK

Proof: Given V^* , need to define Sim^*

Hard case: what if $V^*(R)$ depends on R ?

Zero-Knowledge Proof for QR

$\text{Sim}^*(N, y):$

$(b, R, z) \leftarrow \text{Sim}(N, y)$

If $b^* = b$:

Else, "rewind"

Claim: $y \in \text{QR}_N$

$V^*(N, y; \rho)$

R

b^*

z

Randomness of V^*
(sampled once)

$$\text{Sim}^*(N, y) = ((N, y), \rho, R, z)$$

From final execution
of the loop

Zero-Knowledge Proof for QR

$\text{Sim}^*(N, y):$

$(b, R, z) \leftarrow \text{Sim}(N, y)$

If $b^* = b$:

Else, "rewind"

Claim: $y \in \text{QR}_N$

$V^*(N, y; \rho)$

R

b^*

z

Claim: Sim^* terminates within n attempts with probability $1 - 2^{-n}$

Proof: in one attempt, we have:

$$\Pr[b^* = b] = 1/2$$

Why? Because $b^* = V^*(N, y, R; \rho)$ and R is independent of b !

Zero-Knowledge Proof for QR

$\text{Sim}^*(N, y):$

$(b, R, z) \leftarrow \text{Sim}(N, y)$

If $b^* = b$:

Else, "rewind"

Claim: $y \in \text{QR}_N$

$V^*(N, y; \rho)$

R

b^*

z

Claim: Output of Sim^* identical to $\text{view}_{V^*}(N, y, w)$

Proof: Output of Sim^* is $((N, y), \rho, R, z)$, where

$$(b, R, z) \leftarrow \text{Sim}(N, y) \mid b = b^*$$

Zero-Knowledge Proof for QR

$\text{Sim}^*(N, y)$:

$(b, R, z) \leftarrow \text{Sim}(N, y)$

If $b^* = b$:

Else, "rewind"

Claim: $y \in \text{QR}_N$

$V^*(N, y; \rho)$

R

b^*

z

Claim: Output of Sim^* identical to $\text{view}_{V^*}(N, y, w)$

Proof: Output of Sim^* is $((N, y), \rho, R, z)$, where

$$z = s \leftarrow \mathbb{Z}_N^\times$$

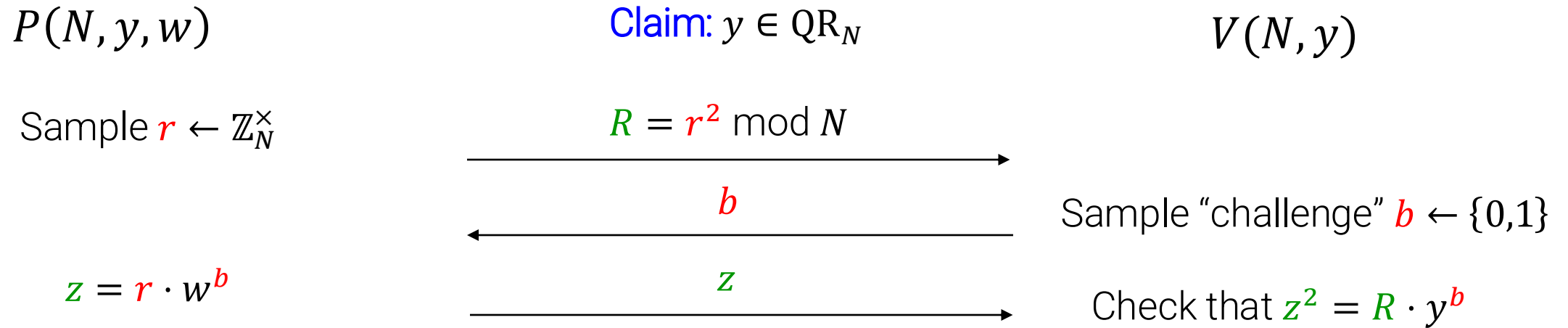
$$R = s^2 \cdot y^{-b^*}$$

$\equiv$

$$z = r \cdot y^{b^*}$$

$$R = r^2$$

Zero-Knowledge Proof for QR



Summary:

- Prover message defined **two claims** (R is a square, $R \cdot y$ is a square)
- If **both claims** are true, y must be a square
- Proof of each **individual claim** can be simulated
- Simulate V^* by guessing **which b^* they will send**, and **rewinding** if wrong

Zero-Knowledge Proofs

Theorem (we just proved):

The language L_{QR} has a **perfect zero knowledge** (PZK) proof system.

- PZK: Simulation indistinguishability is perfect

Question: for which languages are PZK proofs possible?

- Factoring: unknown (may also depend on the formulation of the language)
- 3-coloring: unknown but **strong evidence against**

Next: **computational** zero knowledge